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([redacted])
Bachelor of Computer Applications (BCA) – Third Year
Annual Examination 2026

COMPUTER GRAPHICS
[Paper Code: NP-31]

Max. Marks: 80 | Time: 3 Hours

Pass Marks: 28 (35%) | Date of Exam: _________

Instructions to Candidates:

1. Attempt ALL questions from Section A. Each question carries 2 marks.
2. Attempt any FIVE questions from Section B. Each question carries 6 marks.
3. Attempt any THREE questions from Section C. Each question carries 10 marks.
4. Draw neat diagrams wherever necessary.
5. Use of scientific calculator is **not** permitted.

SECTION – A (Objective / Very Short Answer) — 10 × 2 = 20 Marks

Attempt ALL questions. Each question carries 2 marks.

Q.1. What does DDA stand for in the context of line drawing algorithms? [2]

Q.2. State the decision parameter condition used in Bresenham's Line Drawing Algorithm. [2]

Q.3. Define resolution of a display device. How does it affect image quality? [2]

Q.4. What is a Region Code in Cohen-Sutherland clipping? Write the four-bit codes for all 9 regions. [2]

Q.5. Differentiate between Raster Scan and Random Scan display systems. (Any two points) [2]

Q.6. What is the homogeneous coordinate representation used for in 2D transformations? [2]

Q.7. Define shearing transformation. Give its matrix representation for X-direction shear. [2]

Q.8. What is GIMP? List any three features of GIMP. [2]

Q.9. Define a viewport and a window in the context of computer graphics. [2]

Q.10. What does HSV stand for in color models? Briefly explain its three components. [2]

SECTION – B (Short Answer Questions) — 5 × 6 = 30 Marks

Attempt any FIVE questions. Each question carries 6 marks.

Q.11. Using the DDA line drawing algorithm, rasterize a line from point A(–1, –4) to B(5, 6). Show all steps in tabular form. [6]

Q.12. Explain the Midpoint Circle Drawing Algorithm with a step-by-step example for radius r = 5. List the symmetry property used. [6]

Q.13. Explain the concept of 2D Composite Transformations. Derive the composite matrix for: (i) Translation followed by Rotation, and (ii) Scaling about an arbitrary point. [6]

Q.14. Apply Cohen–Sutherland line clipping algorithm to clip the line P1(10, 30)–P2(80, 90) against the window A(20,20), B(90,20), C(90,70), D(20,70). Show region codes and all clipping steps. [6]

Q.15. Explain Window-to-Viewport coordinate transformation. Derive the transformation equations and illustrate with a suitable diagram. [6]

Q.16. Describe any three selection/drawing tools of GIMP with their purpose and usage. Also explain how to scale an image and increase canvas size in GIMP. [6]

Q.17. Explain Liang–Barsky Line Clipping algorithm. How is it more efficient than Cohen–Sutherland? Illustrate with an example. [6]

Q.18. Explain Flat Panel Display technology. Compare Plasma Panel Display and Liquid Crystal Display (LCD) in terms of working principle, advantages, and disadvantages. [6]

SECTION – C (Long Answer Questions) — 3 × 10 = 30 Marks

Attempt any THREE questions. Each question carries 10 marks.

Q.19. Explain Bresenham's Line Drawing Algorithm in detail. Derive the decision parameter and demonstrate the algorithm by rasterizing a line from (2, 3) to (9, 8). Compare Bresenham's algorithm with DDA and state its advantages. [10]

Q.20. Answer the following: [10]

(a) Perform a 45° clockwise rotation of the triangle A(2,1), B(5,1), C(5,6) about the origin. Give the matrix representation and the resulting coordinates. [5]

(b) Scale the object with vertices A(2,1), B(2,3), C(4,2), D(4,4) with scale factors Sx = 2, Sy = 3 about the fixed point (7,7). Show all steps. [5]

Q.21. Answer the following: [10]

(a) Explain the Sutherland–Hodgeman Polygon Clipping algorithm with a diagram and a worked example. State where it fails. [5]

(b) Explain the Weiler–Atherton Polygon Clipping algorithm. How does it handle concave polygons better than Sutherland–Hodgeman? Illustrate with a diagram. [5]

Q.22. Answer the following: [10]

(a) Define CRT (Cathode Ray Tube). Explain its basic design and working principle with a neat diagram. Differentiate between emissive and non-emissive displays. [5]

(b) Explain the Midpoint Ellipse Drawing Algorithm with step-by-step calculations for semi-axes rx = 8, ry = 6. Write the properties of an ellipse used in the algorithm. [5]

Q.23. Answer the following: [10]

(a) What are layers in GIMP? Explain the various operations that can be performed on layers (add, delete, merge, align, link). How can you create an ID Card design using GIMP? Outline the steps. [5]

(b) Explain the concept of 3D transformations. Derive the transformation matrices for Translation, Rotation about X-axis, Y-axis, and Z-axis, and Scaling in 3D. [5]

★ HIGHLY LIKELY / IMPORTANT TOPICS (Based on Previous Year Analysis)

Unit	Important Topics / Questions (Most Frequently Asked)
Unit 1 **Display Devices**	• CRT structure & working principle • Raster vs Random Scan display • Flat panel display (LCD, Plasma) • Shadow mask & Beam penetration method • Hard copy devices
Unit 2 **Line & Curve Algorithms**	• DDA algorithm with numerical example ★ (asked every year) • Bresenham's line drawing algorithm ★★ (most important) • Midpoint circle algorithm with example ★★ • Midpoint ellipse algorithm • Character generation methods
Unit 3 **2D Transformations**	• Matrix representation of Translation, Rotation, Scaling ★★ • Shearing and Reflection transformations • Composite transformations (most important long question) • Rotation about arbitrary point • Homogeneous coordinates
Unit 4 **Clipping Algorithms**	• Cohen-Sutherland Line Clipping ★★ (numerical always asked) • Liang-Barsky algorithm • Sutherland-Hodgeman Polygon Clipping ★ • Weiler-Atherton polygon clipping • Window to Viewport transformation ★
Unit 5 **GIMP & Color**	• GIMP tools (selection, transform) ★ • Layers in GIMP – operations • ID card design using GIMP • HSV and RGB color models • Gamma correction, Gaussian blur

Note: *This is a* ***DEMO / PRACTICE PAPER*** *prepared based on analysis of DAVV BCA 3rd Year Computer Graphics previous year question banks and syllabus (Annual Pattern). ★★ = Extremely likely to appear ★ = Likely to appear. Always refer to your official syllabus and university-issued model papers for final preparation.*